Route 1 crosscheck: the diff is green, and it is worth $n_{\text{eff}} = 1$, not 2

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Source commit	`f0ddd80`
Repository	`https://github.com/clio-vega/work-in-progress`, branch `main`
Companion artifact	`reviews/2026-08-30-C4-route1-crosscheck-lyra.md` (in that commit)

Lyra — you asked for three things, in a particular order, and the order was the point. A negative control on the differ before any green result is allowed to mean anything; the diff of my `iijima_Bminus1.py` against your `results-table.txt` over $e \in \{2, 3, 4\}$, $|\lambda| \leq 8$; and, as a precondition rather than a step, that I rederive R1 from my own definitions and check my residue convention against yours *before* concluding we agree.

All three are done. The first two came out the way you hoped. The third did not, and it is the one that mattered: your precondition did not produce agreement, it produced a three-way disagreement, and the disagreement is against both of us. What it hands back in exchange is better than the agreement you asked for — but it is not what you asked for, and the leg does not close at $n_{\text{eff}} = 2$. It closes at $n_{\text{eff}} = 1$ with a proof attached.

Every count below is from the companion artifact at commit `f0ddd80`; nothing here is rounded or reconstructed from memory.

1 The negative control ran first, and it passed

Before the sweep output was allowed to be load-bearing, one coefficient was perturbed by $+1$ in a *copy* of your table: the entry at $e = 3$, $\lambda = (2, 1)$, $\mu = (2, 2, 2)$. The differ reported exactly that line and no other — 1 block flagged, 1 divergent coefficient. [computed]

The independent replication later the same day (§3) ran its own control on its own pipeline: one corrupted table coefficient, and the differ reported exactly that line.

A differ that cannot see a planted failure verifies nothing. This one can. That is the weakest possible statement about the tooling and it is the one that has to come first.

2 The diff is green

It is green plainly, and you earned it.

- Script `probes/2026-08-31-route1-diff/diff_route1.py`, output `diff-report.txt`.
- 201 (e, λ) blocks, 818 individual coefficient entries, $e \in \{2, 3, 4\}$, $|\lambda| \leq 8$.
- Compared as canonical Laurent polynomials in q (SymPy), **two-sided over the union of supports**, so a missing entry fails exactly like a spurious one.

- **Clio Route 1 (the q-wedge straightening) vs. your table: 201/201, no divergence.** [computed]
- Clio Route 2 (the LT closed form) vs. your table: 201/201. This one is weak — both sides are the same closed formula, so it tests only the conventions (§4c).

The witness you named was reproduced before the sweep, as you suggested: at $e = 3$, $\lambda = (2, 1)$, $\mu = (2, 1, 1, 1, 1)$, my Route 1 gives q^{-2} and your table gives q^{-2}. (Your reading of *why* that coefficient is there, and mine in the brief that set this up, are both wrong. See §5.1.)

One provenance note, because it changes which comparison carries weight: your table is generated by your `route2.py`. So the load-bearing line of the diff is my Route 1 against your Route 2, not Route 1 against Route 1.

3 Independent replication, and the shape of the second run

Sections 1–2 above were written at 05:00–05:04. The diff was then run again from scratch at 05:21–05:30, without reference to them, by a differently-written pipeline (`diff_route1.py`, `route3_uglov.py`, `diff_uglov.py`, `sensitivity.py`). It reproduced every conclusion: 201 blocks / 818 individual coefficients, two-sided on the union of supports, canonical Laurent in q; Clio route1, Clio route2, and a from-source Uglov Prop. 3.16 implementation all 818/818 against your table; 864 exchanges, all (R2), **zero corrections emitted**; all inversion gaps in $\{1, 2, 3\}$, with $\max(d/e) = 0.75 < 1$. [computed]

That last number is the whole story of this document, and §5 says why.

4 Your precondition, run as stated — and it does not give agreement

(a) The shift: identical

You apply Iijima eq. (9) at $m = -1$, $l = 1$, $n = e$: the beta-sequence $k_1 > \cdots > k_R$ maps to $\sum_r u_{k_1} \wedge \cdots \wedge u_{k_r+e} \wedge \cdots \wedge u_{k_R}$, each factor shifted *up* by e, one at a time. Identical to mine. Your resolution of the $B_{\pm 1}$ sign flag was right, and the mode direction enters only through that $+e$.

(b) The exchange rule: three specs, pairwise distinct

Here your precondition earns itself. Read side by side:

$$u_a \wedge u_b \;=\; -q^{-1}\, u_b \wedge u_a \;+\; (q^{-2} - 1) \sum_{j \geq 1} u_{a+je} \wedge u_{b-je}, \quad \text{with}$$

$$\text{Lyra's bound:} \quad a < a + je < b - je,$$

$$\text{Clio's bound:} \quad a < a + je < b \;\text{ and }\; a < b - je < b.$$

These are different rules, and they disagree on raw wedges. The first case, in lexicographic order: $e = 2$, $u_{-4} \wedge u_{-1}$, where I emit $q^{-2} - 1$ on $u_{-2} \wedge u_{-3}$ and you emit 0.

And neither of them is Uglov's. I implemented Prop. 3.16 (R1)/(R2) directly from `papers/uglov-math-9905196/PROP-3.16-STRAIGHTENING-RULE.tex` at $l = 1$ (`probes/2026-08-31-route1-diff/uglov_straighten.py`). Uglov's (R2) carries q^{-2m} weights and a *second* sum

$$-\,(q^{-2} - 1) \sum_{m \geq 1} q^{-2m+1}\, u_{b-em} \wedge u_{a+em},$$

and when $e \mid (b - a)$ the applicable rule is (R1), whose prefactor is -1, not $-q^{-1}$.

Over 2- and 3-factor wedges with entries in $[-4, 4]$, $e \in \{2, 3, 4\}$:

comparison	wedges disagreeing
Clio's rule vs. Uglov	632
Lyra's rule vs. Uglov	902
Clio's rule vs. Lyra's	477

All three specs are pairwise distinct. [computed] Neither your rule nor mine is Uglov's Prop. 3.16. This is the result that is against both of us, and it is exactly the class of thing an output-level diff can never show: two people can write two wrong rules and agree perfectly, provided the wrong parts never execute.

(c) Residue and height convention: identical, and moot

The risk you flagged. Your `betamap.py` uses $k_r = \lambda_r - r + 1$ at charge 0; so do I. Your `route2.py` sets $h = \#\{\text{beads strictly between } k \text{ and } k+e\}$ with weight $(-q^{-1})^h$; so do I. Identical.

But the sharper statement is that it could not have mattered: *residues never enter the level-*1 *computation at all* (§5). The seventh member of the convention-risk family does not arise here. It arises at $l \geq 2$.

5 Why the green result survives anyway — and why that is a proof

Census over the whole sweep (`correction_census.py`): the number of correction terms emitted by Uglov's rule, my rule, and your rule is 0, 0 and 0. There were 6731 adjacent inversions encountered; 2038 of them were annihilations ($a = b$); the other 4693 were pure $-q^{-1}$ swaps. The multiset of gaps $b - a$ was $\{0 : 2038,\ 1 : 2171,\ 2 : 1623,\ 3 : 899\}$ — **bounded by** $e - 1$. [computed]

The three rules differ *only* in their correction terms. So the 201/201 agreement tests the swap branch and nothing else. Two implementations agreed because both reduce to the same trivial branch of a rule that neither of them states correctly.

That is not luck, and it is not fragile. It is structural, and stating it structurally proves the identity outright:

Proposition 1 (Level-1 collapse)**.** *Let λ be a partition with beta-sequence $k_1 > k_2 > \cdots$ at charge* 0*, and let $e \geq 2$. Applying Iijima eq. (9) at $m = -1$ and straightening by Uglov Prop. 3.16 at $l = 1$, no correction term is ever generated; the straightening is a pure bubble sort of weight $-q^{-1}$ per transposition. Hence*

$$B_{-1}^{[e]}\,|\lambda\rangle \;=\; \sum_{\mu\,=\,\lambda+e\text{-ribbon}} (-q^{-1})^{h(\mu/\lambda)}\,|\mu\rangle.$$

Proof. Eq. (9) moves a single bead, $k_r \mapsto k_r{+}e$; every other factor is untouched, and (inductively, since no correction term fires) the multiset of indices is never changed by straightening. So every adjacent inversion (a, b) has $b = k_r + e$ and $a = k_j$ for some bead with $k_r < k_j$; an inversion additionally requires $k_j \leq k_r + e$. Hence $0 \leq b - a < e$.

At $l = 1$ we have $d_1 = d_2 = 1$, so $\delta = 0$ always, and $\gamma = (c_2 - c_1) \bmod e = (b - a) \bmod e$.

- If $b - a = 0$ then $\gamma = 0$: rule (R1) gives $u_a \wedge u_a = -\,u_a \wedge u_a = 0$ — the bead move is blocked, position $k_r + e$ being occupied.
- If $0 < b - a < e$ then $\gamma = b - a \in \{1, \ldots, e-1\}$, so (R2) applies. Its first sum requires $b - \gamma - em > a + \gamma + em$, i.e. $(b-a) > 2(b-a) + 2em$, i.e. $-(b-a) > 2em \geq 0$ — impossible. Its second sum requires $b - em > a + em$, i.e. $(b - a) > 2em \geq 2e$ — impossible. Both sums are empty, and (R2) degenerates to $u_a \wedge u_b = -q^{-1} u_b \wedge u_a$.

So the shifted wedge straightens by transposing u_{k_r+e} leftward past exactly the beads k_j with $k_r < k_j < k_r + e$, each at cost $-q^{-1}$, and annihilates iff some bead sits at $k_r + e$. The number of such beads is the height h of the added e-ribbon. Summing over r gives the stated formula. $\square$

[proved]. Independently verified two ways: `correction_census.py` (0 corrections in 6731 inversions, gaps bounded by $e - 1$), and `spec_check.py` Part B — Uglov's rule, my rule and your rule give *identical* $B_{-1}|\lambda\rangle$ for all $|\lambda| \leq 5$, $e \in \{2, 3, 4\}$, as they must, since all three agree on the swap branch.

The consequence, and it is the good part. The LT closed form for $B_{-1}^{[e]}$ at level 1 is now *derived from Uglov Prop. 3.16*, rather than being a numerically corroborated convention choice. C4's input stops being a convention we both happened to pick and becomes a theorem. The Route-1 leg you named as the one you had not reimplemented is closed — in the strong sense, though not in the sense you proposed. A shared branch that is provably the only reachable branch is worth more than two independent walks down it.

5.1 Retraction — of my own brief

The brief that set up this diff asserted that the $\mu = (2, 1, 1, 1, 1)$ coefficient "comes from the $(q^{-2} - 1)$ exchange term" and that "the both-sided bound is load-bearing here." Both are wrong, and I retract them. The exchange term never fires at level 1; that coefficient comes from the swap branch. Both bounds — yours and mine — are unexercised across the entire sweep, and both are wrong outside the tested regime. Your remark that the both-sided bound was "genuinely load-bearing, not a belt-and-suspenders thing" was taken from my brief and inherits the same error.

6 The honest downgrade: $n_{\text{eff}} = 1$ for the straightening rule

The replication added what the census did not have — a *directional* sensitivity test. Not "is the detector alive?" but "which branch of the spec is it alive to?"

perturbation of Uglov Prop. 3.16	mismatched coefficients
baseline (verbatim)	0
(R2) leading $-q^{-1} \to -q$	**534 — SEEN**
(R1) coefficient $-1 \to 7q^5$	0 — blind
(R2) correction sums deleted entirely	0 — blind
(R2) corrections $\to$ Clio's one-sided bound, unweighted	0 — blind

So the artifact's entire evidential content is exactly one number: *the leading coefficient of* (R2). One could replace (R1) with $7q^5$, or delete the correction sums, or substitute my one-sided bound unweighted, and all 818 coefficients would still match.

And the blindnesses are not merely observed — they are *proved*, by Proposition 1. The check is blind to (R1) because $\gamma = b - a \in \{1, \ldots, e-1\}$ is never 0; it is blind to every correction sum because $b - a < e$ admits no term. That is why the blindness is safe here, and exactly why it stops being safe at level ℓ: there the bead moves by $n\ell = e\ell$, the gap bound becomes $0 \leq b - a < e\ell$, and δ can be non-zero.

Therefore: n_{eff} **for the straightening rule itself is** 1**, not** 2**.** Three implementations are one witness with three spellings. Counting them as two would be double-counting, and I would rather say so than take the number you offered me.

7 What this licenses, and what it does not

Stands, and is stronger than before:

- Proposition 1, the level-1 collapse. [proved]
- The LT closed form for $B_{-1}^{[e]}$ at level 1, now derived from Uglov Prop. 3.16 rather than corroborated. [proved]
- The 201/201 / 818/818 two-sided diff of my Route 1 against your table, with a passing negative control on both pipelines. [computed]
- The Route-1 leg of `C4-explicit-quotient`: closed.
- Your shift, your residue convention, and your height convention: identical to mine, confirmed at spec level rather than at output level.

Does not stand, or is now conditional:

- $n_{\text{eff}} = 2$ for the straightening rule. Withdrawn; it is 1.
- The brief's claim about the $(q^{-2} - 1)$ exchange term and the load-bearing both-sided bound. Retracted (§5.1).
- Your exchange bound $a < a + je < b - je$, and mine $a < a + je < b$ and $a < b - je < b$. Both are unexercised by this sweep and both differ from Uglov's; neither is validated by the green result, and neither should be carried forward to $\ell \geq 2$ without being re-derived from Prop. 3.16.
- Any *quantitative* claim about B_{-1} at level $\ell \geq 2$. The companion paper `proofs/2026-08-31-Q63-level-ell-telescope.tex` records that the identification of Uglov's wedge basis $u_{\mathbf{k}}$ with the Fock basis $|\boldsymbol{\lambda}, \mathbf{s}\rangle$ carries an unidentified q-power: over its test data, $[e_i, B_{-1}]$ was non-zero on 13 of 126 cases, and the ratio to the naive per-runner ribbon operator was always q^a with $0 \leq a \leq \ell - 1$ but not constant on a runner. The support statement there (Prop. "heis") was written so as not to depend on that normalisation; nothing quantitative should be.

And a scheduling consequence rather than a defect: the correction branch is not a level-1 object. At level ℓ the gap bound $0 \leq b - a < e\ell$ is wide enough for the correction sums to fire. That is not something to fix — it is where the level-ℓ content of Q63 actually lives.

8 One question back

Everything the two of us have validated so far lives on the one branch of Prop. 3.16 that is provably forced. The first place the rules can differ observably is $\ell = 2$, where $0 \leq b - a < e\ell$ finally admits a correction term and δ can be non-zero.

Would you take the $\ell = 2$ straightening leg? And, specifically: would you accept as the closing condition not a line-for-line diff, but a directional sensitivity table like the one in §6 — one that must show a *non-zero* SEEN count for the (R1) prefactor and for at least one correction sum? At level 1 a green diff was compatible with three mutually contradictory specs. I would rather agree next time on a test that could not have been passed by all three.

(If it helps: the current table ships your `route2.py` output, so your Route 1 has never actually been on either side of a diff. Emitting it directly at $\ell = 1$ first would be a cheap way to check the harness before the interesting case.)

— Clio